

Pune Weather Pattern Analysis

Project Documentation Report

Project 2 • One Step Forward from Sales Trend Analysis
Beginner Data Analytics • Tool: Google Sheets • Dataset: Pune, India • 2024

1. Project Overview

This is the second project in a beginner data analytics learning journey. It analyses 73 years of daily weather data recorded in Pune, Maharashtra, India — from 1951 to 2024. The goal is to identify seasonal patterns in temperature and rainfall, understand the monsoon cycle, and detect any long-term temperature trends using Google Sheets.

This project introduces five new Google Sheets skills beyond what was learned in the previous Sales Trend Visualisation project, making it a deliberate step forward in data analytics ability.

2. Objectives

- Identify which months are the hottest and coolest in Pune across 73 years of records
- Visualise the monsoon rainfall pattern by month using a bar chart
- Detect whether Pune's average temperature has changed over the decades using a yearly trend chart
- Apply conditional formatting to create a colour-coded temperature heatmap without using a chart
- Use sparklines to show mini trends inside a summary table
- Practice new Google Sheets skills: Freeze Rows, AVERAGEIFS, Combo Chart, Conditional Formatting, and Sparklines

3. Dataset Description

The dataset used is `pune.csv` — a historical daily weather record for Pune, India, sourced from meteorological records.

Dataset at a glance

Property	Detail
File name	pune.csv
Total rows	26,803 daily records
Time span	1st January 1951 to June 2024 (73 years)

Location	Pune, Maharashtra, India
Data source	Meteorological historical weather records

Columns in the dataset

Column	Example Value	Description
Date	01-01-1951	Date of the record in DD-MM-YYYY format
Rain	8.69	Daily rainfall in millimetres (mm). Value of 0 means no rain.
Temp Max	31.8	Maximum temperature recorded that day in degrees Celsius
Temp Min	24.4	Minimum temperature recorded that day in degrees Celsius

Note: The dataset contains some rows where Temp Max or Temp Min shows ----- (missing data). These were handled by using the AVERAGEIFS formula with a condition to exclude them from calculations.

4. Tools Used

Tool	Purpose
Google Sheets	All data import, cleaning, formulas, summary tables, charts, and formatting
pune.csv	Source dataset — 26,803 rows of daily Pune weather data
GitHub	Project file hosting and submission repository
Google Drive	Cloud storage for the spreadsheet and downloaded output files

No programming or coding was used in this project. All analysis was performed using Google Sheets built-in features only.

5. New Skills Learned in This Project

This project was designed as a step forward from the previous Sales Trend Visualisation project. The following five new Google Sheets skills were applied for the first time:

1

Freeze Rows

Locks the header row so column names stay visible while scrolling through 26,000+ rows. Applied via View > Freeze > 1 row on the Raw Data sheet.

2

AVERAGEIFS Formula

Calculates an average with multiple conditions. Used to find average monthly temperature while skipping rows with missing ----- values. More powerful than the AVERAGEIF formula (single condition) used in simpler projects.

3

Combo Chart (Dual Axis)

A single chart that combines a line and column bars, each with its own Y-axis scale. Used to show yearly temperature (line, right axis in degrees C) and yearly rainfall (bars, left axis in mm) together without one series being too small to see.

4

Conditional Formatting — Colour Scale

Automatically colours cells from white to yellow to red based on their value, creating a visual heatmap without inserting any chart. Applied to the Avg Temp Max column and Total Rain column of the Monthly Summary sheet.

5

Sparklines

A mini chart that fits inside a single cell. Used to show the 12-month temperature trend and rainfall trend as tiny inline visuals inside the Monthly Summary table. Created using the SPARKLINE() formula.

6. Steps Performed

Step 1 — Data Import and Sheet Setup

The pune.csv file was uploaded to Google Sheets via File > Import > Upload. The sheet tab was renamed Raw Data. The header row was frozen (View > Freeze > 1 row) so column names stayed visible while scrolling through over 26,000 rows. This was the first application of the Freeze Rows skill.

Step 2 — Adding Helper Columns

Three new columns were added to the Raw Data sheet to extract the month and year from the Date column, since the dates were stored as text (DD-MM-YYYY) rather than date values:

- Column E (Month_Num): =MID(A2,4,2)*1 — extracts the 2-digit month number
- Column F (Year): =RIGHT(A2,4)*1 — extracts the 4-digit year
- Column G (Month_Name): =TEXT(DATE(2000,E2,1),"MMM") — converts month number to short name (Jan, Feb, etc.)

All three formulas were dragged down to row 26,804 to cover the full dataset.

Step 3 — Monthly Summary Sheet

A new sheet tab named Monthly Summary was created with rows for all 12 months. Three summary formulas were applied:

- Avg Temp Max (column C): AVERAGEIFS formula averaging Temp Max for each month number, excluding ----- values
- Avg Temp Min (column D): Same formula applied to Temp Min column
- Total Rain (column E): SUMIF formula totalling rainfall for each month across all 73 years, then divided by 73 to get average annual monthly rainfall in mm

Sparklines were added in columns F and G using the SPARKLINE() formula — a red line chart for temperature and a blue line chart for rainfall, each appearing inside a single cell.

Step 4 — Yearly Summary Sheet

A new sheet tab named Yearly Summary was created with one row per year from 1951 to 2024. AVERAGEIFS formulas calculated average Temp Max, Temp Min, and total rainfall for each year.

Step 5 — Conditional Formatting Heatmap

The Avg Temp Max column (C2:C13) in Monthly Summary was selected and conditional formatting was applied via Format > Conditional formatting > Color scale. The scale was set to white (low) > yellow (mid) > red (high), automatically colouring April and May dark red as the hottest months. The same technique was applied to Total Rain with a white to dark blue scale, making the monsoon months visually obvious.

Step 6 — Chart Creation

Four visualisations were produced:

- Chart 1: Line chart of monthly average Temp Max and Temp Min — showing the temperature cycle across the 12 months
- Chart 2: Bar chart of average monthly rainfall — clearly showing the June to September monsoon spike
- Chart 3: Combo chart of yearly average temperature (line, right axis) and yearly total rainfall (columns, left axis) — showing trends across 1951 to 2024
- Chart 4: Screenshot of the conditional formatting heatmap table — showing temperature and rainfall colour scales across all 12 months

Step 7 — Export and Submission

The spreadsheet was downloaded as an Excel .xlsx file. All chart images were downloaded as PNG files. Screenshots of the Monthly Summary heatmap and sparklines were captured. All files were organised into folders and uploaded to GitHub.

7. Charts and Visualisations

Chart 1 — Monthly Average Temperature (Line Chart)

A dual-line chart showing average Temp Max and Temp Min for each of the 12 months. Reveals that Pune is hottest in April and May (around 35°C) and coolest in December and January (around 15°C minimum). Temperature dips during June to August due to monsoon cloud cover.

File: charts/chart1_monthly_temp.png

Chart 2 — Monthly Rainfall (Bar Chart)

A bar chart showing average total rainfall per month. The monsoon pattern is extremely clear: June, July, and August dominate with July being the peak month. The dry months (October to May) show near-zero bars by comparison.

File: charts/chart2_monthly_rain.png

Chart 3 — Yearly Temperature and Rainfall Trend (Combo Chart)

A dual-axis combo chart showing average Temp Max as a line (right Y-axis, in degrees C) and total annual rainfall as bars (left Y-axis, in mm) from 1951 to 2024. This chart required setting Avg Temp Max to the right axis so the temperature line was visible above the rainfall bars. This technique is called a dual axis chart and is used when two series have very different value scales.

File: charts/chart3_yearly_trend.png

Chart 4 — Conditional Formatting Heatmap (Screenshot)

The Monthly Summary table with colour-scale conditional formatting applied to both Avg Temp Max (white-yellow-red) and Total Rain (white-light blue-dark blue) columns, plus

sparkline mini charts in the last two columns. This is a screenshot rather than an exported chart since conditional formatting is a table feature, not a chart.

File: charts/chart4_heatmap.png

8. Key Findings

- April and May are the hottest months in Pune with average maximum temperatures around 35°C
- December and January are the coolest months with minimum temperatures dropping to around 15°C
- The monsoon season (June to September) brings the vast majority of annual rainfall, with July being the peak month
- During the monsoon months, maximum temperatures actually drop to around 28°C due to cloud cover, despite being summer months
- The dry season from October to May receives very little rainfall, with January and February being almost completely dry
- The 73-year yearly trend chart reveals gradual changes in average temperature and variability in annual rainfall across the decades

9. Challenges Faced

- The date column was stored as text (DD-MM-YYYY) rather than a proper date format, so MID and RIGHT formulas were needed to extract the month and year numbers instead of using built-in date functions
- The dataset contained ----- values for missing temperature records in some rows. These had to be excluded using the additional condition in AVERAGEIFS to avoid errors
- The combo chart initially showed the temperature line completely flat at the bottom because the rainfall values (up to 5000mm) were far larger than temperature values (30-35 degrees). This was fixed by assigning Avg Max Temp to the right axis, creating a dual axis chart
- Google Sheets calls vertical bars Columns not Bars inside the combo chart Series settings, which caused initial confusion when looking for the correct chart type option

10. GitHub Repository Structure

File Path	Description
dataset/pune.csv	Raw dataset — 26,803 daily weather records (1951–2024)
analysis/pune_weather_analysis.xlsx	Google Sheets export with Raw Data, Monthly Summary, and Yearly Summary tabs
charts/chart1_monthly_temp.png	Line chart — Monthly average temperature (Temp Max & Min)

charts/chart2_monthly_rain.png	Bar chart — Monthly average rainfall showing monsoon pattern
charts/chart3_yearly_trend.png	Combo chart — Yearly temperature trend and rainfall (1951–2024)
charts/chart4_heatmap.png	Screenshot — Conditional formatting heatmap with sparklines
screenshots/output_screenshot.png	Screenshot of Google Sheets showing summary table and charts
docs/documentation.pdf	This documentation report
README.md	Project description and overview on GitHub

11. What I Learned Compared to Project 1

Skill	Project 1 (Sales)	Project 2 (Weather)
Data import	CSV upload	CSV upload + helper columns
Formula	SUMIF	AVERAGEIFS (multiple conditions)
Grouping	Pivot Table	Manual summary with formulas
Charts	Bar chart, Pie chart	Line, Bar, Combo (dual axis)
Visual design	Basic chart colours	Conditional formatting heatmap + Sparklines
Navigation	No freeze	Freeze rows for large dataset

12. Conclusion

This project successfully analysed 73 years of Pune weather data using only Google Sheets. The analysis confirmed clear seasonal patterns: Pune has a hot dry summer peaking in April-May, a heavy monsoon from June to September, and a cool dry winter from November to February.

Five new Google Sheets skills were applied for the first time in this project: Freeze Rows, AVERAGEIFS, Combo Chart with dual axis, Conditional Formatting colour scale, and

Sparklines. Each skill solved a real problem encountered during the analysis, making the learning practical and meaningful.

The dual axis combo chart challenge — where the temperature line was initially invisible because rainfall values were much larger — was the most valuable learning moment, introducing the concept of axis scaling in data visualisation.

This project builds a strong foundation for future analytics work involving time-series data, climate analysis, and dashboard-style reporting in Google Sheets.

End of Documentation

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